



Field	Bit(s)	Init Val	Description
PSR_type17	17	x	Reserved.
PSR_type18	18	x	Reserved.
Reserved	28:19	x	Reserved.
RQPL	31:29	x	Defines the number of bits to use for RSS redirection of packets dedicated to this pool. Valid values are zero, 0001b and 0010b. The default value should be 0010b, meaning that up to 4 queues can be enabled for this pool. A value of 0001b means that up to 2 queues can be enabled for this pool. A value of zero means that all the traffic of the pool is sent to queue 0 of the pool. This field is used only if MRQC.MRQE equals 1010b or 1011b.

8.2.3.7.5 Receive Checksum Control — RXCSUM (0x05000; RW)

Field	Bit(s)	Init Val	Description
Reserved	11:0	0x0	Reserved.
IPPCSE	12	0b	IP Payload Checksum Enable.
PCSD	13	0b	RSS/Fragment Checksum Status Selection. When set to 1b, the extended descriptor write back has the RSS field. When set to 0b, it contains the fragment checksum.
Reserved	31:14	0x0	Reserved.

The Receive Checksum Control register controls the receive checksum offloading features of the 82599. The 82599 supports the offloading of three receive checksum calculations: the fragment checksum, the IP header checksum, and the TCP/UDP checksum.

PCSD: The Fragment Checksum and IP Identification fields are mutually exclusive with the RSS hash. Only one of the two options is reported in the Rx descriptor. The RXCSUM.PCSD affect is shown in [Table 8-7](#).

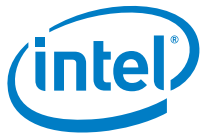
Table 8-7 Checksum Enable/Disable

RXCSUM.PCSD	0b (Checksum Enable)	1b (Checksum Disable)
	Fragment checksum and IP identification are reported in the Rx descriptor.	RSS hash value is reported in the Rx descriptor.

IPPCSE: IPPCSE controls the fragment checksum calculation. As previously noted, the fragment checksum shares the same location as the RSS field. The fragment checksum is reported in the receive descriptor when the RXCSUM.PCSD bit is cleared.

If RXCSUM.IPPCSE is cleared (the default value), the checksum calculation is not done and the value that is reported in the Rx fragment checksum field is 0b.

If RXCSUM.IPPCSE is set, the fragment checksum is aimed to accelerate checksum calculation of fragmented UDP packets. See [Section 7.1.13](#) for a detailed explanation.



This register should only be initialized (written) when the receiver is not enabled (for example, only write this register when RXCTRL.RXEN = 0b).

8.2.3.7.6 Receive Filter Control Register — RFCTL (0x05008; RW)

Field	Bit(s)	Init Val	Description
Reserved	5:0	0x0	Reserved.
RSC_DIS	5	0	RSC Disable. When set, disable RSC for the port by the Rx filter unit. The default value is 0b (RSC feature is enabled).
NFSW_DIS	6	0b	NFS Write disable. Disable filtering of NFS write request headers.
NFSR_DIS	7	0b	NFS Read disable. Disable filtering of NFS read reply headers.
NFS_VER	9:8	00b	NFS version recognized by the hardware. 00b = NFS version 2 01b = NFS version 3 10b = NFS version 4 11b = Reserved for future use
IPv6_dis	10	0b	IPv6 Disable. Disable IPv6 packet filtering Internal use only – should not be set to 1b.
Reserved	11	0b	Reserved, always set to 0b.
Reserved	13:12	00b	Reserved.
IPFRSP_DIS	14	0b	IP Fragment Split Disable When this bit is set the header of IP fragmented packets are not set. Internal use only. Should not be set to 1b.
Reserved	15	0b	Reserved.
Reserved	17:16	00b	Reserved.
Reserved	31:18	0x0	Reserved. Should be written with 0x0 to ensure future compatibility.